



SECURITATEA INFORMAȚIONALĂ 2012

CONFERINȚĂ INTERNAȚIONALĂ,
(ediția a IX-a), 18 iunie 2012

ACADEMIA DE STUDII ECONOMICE DIN MOLDOVA
LABORATORUL DE SECURITATE INFORMAȚIONALĂ

SECURITATEA INFORMAȚIONALĂ 2012

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(ediția a IX-a)

18 iunie 2012

Chișinău – 2012

CZU 004.056(082)=135.1=111=161.1

S 40

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Descrierea CIP a Camerei Naționale a Cărții

„Securitatea informațională 2012”, conf. intern. (2012; Chișinău). Securitatea informațională 2012: Conf. intern. (ed. a 9), 18 iun 2012 / resp. de ed.: S. Ohrimenco. – Ch.: ASEM, 2012. – 78 p. Antetit.: Acad. de Studii Econ. din Moldova, Lab. de Securitate Informațională. – Texte: lb. rom., engl., rusă. – Bibliogr. la sfârșitul art. –100 ex.

ISBN 978-9975-75-499-6.

CD-rom:sd, col.; in container, 19 x 14 x 2 cm. Cerinte de system: Windows 98/2000/XP, 64 Mb hard, VOB Media Player

004.056(082)=135.1=111=161.1

Coordonatorul ediției - prof.univ. dr. hab. **S. Ohrimenco**

© Laboratorul de Securitate Informațională al ASEM

ISBN 978-9975-75-499-6

Laboratorul de Securitate Informațională al ASEM este membru al
Asociației Europene pentru Securitate



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ANOMALY DETECTION 3.0 FOR SNORT®

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Snort® is an open source intrusion detection system based on signature detection. In the paper we present information about the third version of Snort AD – preprocessor designed to log and analyze network traffic information developed by us.

AD preprocessor

Snort is the most popular open source intrusion detection system based on signature detection (see e.g. [2], [14], [18], [19]). The modular construction of Snort allows one to extend its capabilities by creating own pre- or postprocessors and/or plugins. The best-known Snort tools are for instance ACID (see [15]), BASE (see [16]) or SAFE (see [17]).

In our works we develop Snort preprocessor designed to enhance Snort possibilities to monitor, analyze and detect network traffic anomalies using NBAD (Network Behavioral Anomaly Detection) approach (see e.g. [6], [7], [20]). The first version of Anomaly Detection preprocessor (see [21]) for Snort version 2.4x was published in a Master's Thesis [11] in 2006. Next the project has been developed (see e.g. [10], [12], [9], [8]) till the current version 3.0 designed for Snort 2.9.x which periodically, with a given interval, logs information about 29 parameters of the network traffic as a number of TCP/UDP packets sent/received from outside/inside the current IPv4 subnet, www download/upload speed, a number of UDP 53 (DNS) datagrams etc. Values of these parameters are the logs into a file in CSV (Comma Separated Values) format, with header line containing description of each parameter (see Figure 4).

DD-MM-YY, HH:MM:SS, Day of the Week, Time interval [s], TCP summary [number of packet], TCP outgoing [number of packet], TCP incoming [number of packet], TCP from this subnet [number of packet], UDP summary [number of packet], UDP outgoing [number of packet], UDP incoming [number of packet], UDP from this subnet [number of packet], ICMP summary [number of packet], ICMP outgoing [number of packet], ICMP incoming [number of packet], ICMP from this subnet [number of packet], TCP with SYN/ACK [number of packets], WWW outgoing - TCP outgoing to port 80 [number of packet], WWW incoming - TCP incoming from port 80 [number of packet], DNS outgoing - UDP outgoing to port 53 [number of packet], DNS incoming - UDP incoming from port 53 [number of packet], ARP-request [number of packet], ARP-reply [number of packet], Not TCP/IP stacks packet [number of packet], Total [number of packet], TCP upload speed [kBps], TCP download speed [kBps], WWW upload speed [kBps], WWW download speed [kBps], UDP upload speed [kBps], UDP download speed [kBps], DNS upload speed [kBps], DNS download speed [kBps]
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The profile can be generated “manually” or by a Profile Generator using appropriate model based on historic values from the log file. The architecture affords easy implementation of different statistical models of the traffic and usage of different tools (i.e. statistical packets) for building profiles. For easy implementation of adaptive models (which have to generate profile ‘incrementally’ after getting current values of the traffic parameters), the values from the profile file are loaded by AD to the log file after each saving of the current traffic values.

Profile Generator

In the current version of AD the profile generator was designed based on R environment (see:[13]).

The profile generator produces four files:

- File containing predicted pattern (expected future values of parameters) of the network traffic based on the statistical model for a given future time period.
- Profile file containing minimum and maximum values of the parameters (limitations for alert generation).
- File containing calculated values of the model parameters.
- File containing traffic pattern (theoretical values of the parameters) for the past time (the same time as in the log file). This values are used to evaluate the quality of the model.

(See Figure 6), gray solid arrows means saving to and the black dotted ones – reading from the file.

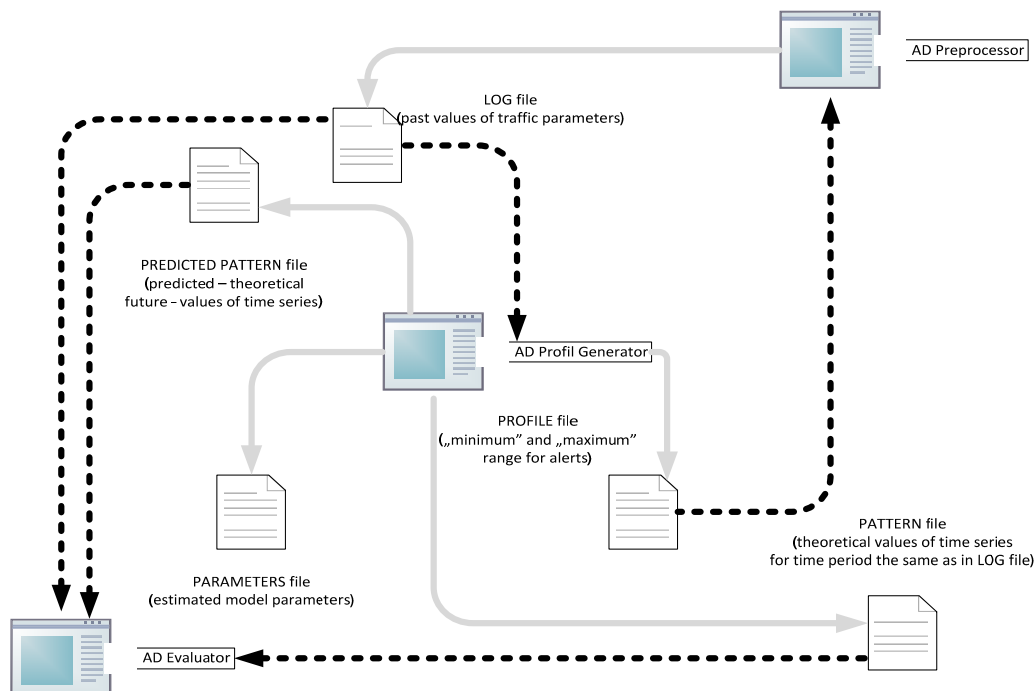


Figure 6. AD Data flow diagram. Source: own research.Evaluator

In the current version the Profile generator can build profiles based on four methods: Moving average, Naïve method, Autoregressive time series model and Holt-Winters model (see e.g. [4], [3], [1], [5], [8]).

The third program in the project (named Evaluator) is designed to compare simply statistic $\frac{MAE}{M}$ for two files (e.g. log file and predicted value profile).

MAE means Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t| = \frac{1}{n} \sum_{t=1}^n |e_t|$$

where $e_t = y_t - \hat{y}_t$ is a model residual in the moment t and M is arithmetic mean.

This way the Evaluator can be used either for checking fit between the model and historical data or between the predicted and real values.

All of the programs can be downloaded for free from the AD project page <http://www.anomalydetection.info>

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КЛАССИФИКАЦИЯ ДАННЫХ КАК АСПЕКТ ИНФОРМАЦИОННОЙ БЕЗОПАСНОСТИ

Дмитрий Дорошев, Ольга Корнеевко

In article aspects of the information safety, concerning classifications of data are considered. Also some problems of subjects of the managing, connected with unwillingness to be engaged in this process are presented.

Практика показывает, что, разрабатывая проекты по информационной безопасности, субъекты хозяйствования часто забывают об одном немаловажном аспекте – классификации данных. В корпоративной сети информация хранится не только в папках файловых хранилищ и файлах офисных приложений. Есть еще масса программного обеспечения корпоративного уровня, ERP-, CRM- и HRM-системы, базы данных. Но и данные, хранящиеся в этих системах, могут быть классифи-